M.Tech Program

Advanced Industry Integrated Programs

Jointly offered by University and LTI MindTree

Graph RAG based Financial Insights

Review 2

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Introduction…. Calibri body -36

Problem

- Annual reports = **Tables + Text + Strategy + Risk**
- Financial information is **distributed across pages**
- Traditional search finds **text**, not relationships
- LLMs may generate **unsupported answers**

Proposed Idea

- Financial Fact Extraction
- Entity & Relationship Extraction
- **Knowledge Graph**
- Multi-Hop Graph Retrieval
- **Ollama or Any API provider**
- Source/Page Grounding

Objective…. Calibri body -36

- Extract **financial facts**
- Identify **business entities**
- Discover **semantic relationships**
- Build a **Neo4j financial knowledge graph**
- Perform **multi-hop reasoning**
- Retrieve **source/page evidence**
- Generate answers using **local model**
- Minimize **hallucination**

Module name – Core Engine

- **Hybrid Retrieval –** Graph + vector retrieval
- **Knowledge Graph –** Connected financial entities
- **Fact Extraction –** Structured financial facts
- **Provenance Tracking –** Source-backed evidence
- **Multi-Hop Reasoning –** Graph traversal insights
- **Answer Generation –** LLM-powered responses
- **Multi-Year Analysis –** Fiscal trend tracking
- **Deterministic Ingestion –** Zero-LLM SEC processing

Methodology/Modeling Plan

- **SEC Ingestion Pipeline (sandbox_engine)**
  - EDGAR index → accession discovery → filing download
  - Zero-LLM parsing → structured tables → Arrow/Parquet checkpoints
  - Schema validation → LadybugDB bulk load with idempotent updates
- **Query Routing & Cold-Start GraphRAG**
  - Entity resolution using ticker, company name, aliases, or symbols
  - Routes queries as KNOWN, COLD_START, or AMBIGUOUS
  - Cold-start: fetch → LLM extraction → graph overlay → hybrid retrieval → grounded synthesis
- **Domain-Agnostic GraphRAG**
  - PDF → chunking → LLM-guided triple extraction
  - Validated graph insertion with idempotent upserts
  - Entity lookup → graph traversal → evidence assembly → answer generation

Data preprocessing & EDA

- **500+** issuers across **10-K, 10-Q, 8-K, DEF 14A** and exhibits
- **Multi-year** financial corpus with normalized fiscal calendars
- **12,000+** financial concepts extracted with alias-based deduplication
- **45,000+** graph relationships validated through integrity checks
- Company identity maintained using **SEC ticker–CIK mappings**
- Evidence lineage tracked for every fact with **source** and **provenance**
- **10-stage checkpoint/recovery** pipeline with retry and backoff mechanisms

Approach/Algorithms/Tools

Technology Purpose	Technology Purpose
LadybugDB Embedded property graph storage	LadybugDB Embedded property graph storage
DuckDB / Arrow / Parquet Columnar processing & checkpoints	DuckDB / Arrow / Parquet Columnar processing & checkpoints
Cypher MERGE Idempotent graph update	Cypher MERGE Idempotent graph update
NetworkX Cold-start overlay graph	NetworkX Cold-start overlay graph
OpenAI / Gemini / NIM LLM extraction & synthesis	OpenAI / Gemini / NIM LLM extraction & synthesis
FiscalCalendar Engine Fiscal period alignment	FiscalCalendar Engine Fiscal period alignment
SSRF-Protected Client Secure SEC data retrieval	SSRF-Protected Client Secure SEC data retrieval
ConceptRegistry Entity resolution & deduplication	ConceptRegistry Entity resolution & deduplication

Project plan/milestones progress

- SEC filing **ingestion** and deterministic **parsing** completed
- **Entity resolution** and **financial concept extraction** implemented
- **Knowledge graph creation** with **45K+ validated relationships** completed
- **Arrow/Parquet checkpointing** and **bulk graph loading** operational
- **Provenance tracking**, **citations**, and **retrieval framework** completed
- **Hybrid GraphRAG** pipeline with **cold-start query support** implemented
- **UI layer** and **external database integration** currently in progress

Pipeline structure

